A Coupled Differential Equation Model of Interspecies Competition of Fungal Species and Moisture Niche Breadth

YongJin Lee, Justin Lee, Chloe Auyeung, Oliver Zhou, Anish Chintalgattu

Summary

Wood decomposition plays an important role in ecosystems and the global carbon cycle because it recycles the carbon stored in plants back into the soil and atmosphere. Fungi are the primary organisms responsible for decomposing wood, but fungal species greatly differ in their growth rates, environmental tolerances, and competitive abilities. Natural environments also experience seasonal and stochastic fluctuations in environmental conditions such as moisture. Because of this variability, it is difficult to determine which fungal traits have the strongest influence on decomposition rates and how fungal communities respond to changing environmental conditions. Understanding how fungal traits interact with environmental variability can help improve predictions of wood decomposition in ecosystems experiencing environmental change.

In this project, we develop a mathematical model to study how fungal species grow, compete, and decompose woody biomass under varying moisture conditions. The model consists of a system of coupled differential equations that track both fungal population abundance and the amount of wood biomass remaining over time. Species-specific decomposition rates depend on hyphal extension rate, which represents fungal growth rate, and moisture tolerance, which represents the ability of a species to grow and function across different moisture conditions. Environmental moisture is incorporated as a time-dependent forcing function that includes both seasonal fluctuations and stochastic variability. By combining biological traits with changing environmental conditions, the model provides a framework for exploring how fungal populations respond dynamically to environmental variability and how these responses influence long-term decomposition processes.

We first analyze a simple single-species model to understand how fungal growth and wood decomposition respond to different moisture conditions. We then extend the model to include interspecies competition between fungi with different traits, such as fast-growing specialists that perform best under narrow environmental conditions and slower-growing generalists that tolerate a wider range of moisture levels. Additional simulations examine the effects of environmental variability, biodiversity, and sudden environmental perturbations on decomposition. These simulations allow us to compare how fungal communities behave under stable and fluctuating environmental conditions, and to identify the traits that most strongly influence decomposition.

Our results show that wood decomposition occurs most efficiently when environmental moisture levels are close to the preferred conditions of the dominant fungal species. Increased environmental variability reduces decomposition efficiency because fungal populations spend less time near their optimal growth conditions. In competitive communities, fast-growing specialists tend to dominate under stable environmental conditions, while moisture-tolerant species become more important when environmental variability increases. Simulations with a greater number of species also demonstrate that fungal biodiversity can stabilize decomposition activity across changing environmental conditions by allowing different species to perform well under different moisture conditions.

Overall, the model highlights how fungal traits, environmental variability, and species interactions work together to impact wood decomposition. These results suggest that maintaining diverse fungal communities may help sustain decomposition processes in ecosystems experiencing increasing environmental variability, since different fungal species can contribute to decomposition under different environmental conditions.

1 Introduction

The carbon cycle involves the movement of carbon throughout the earth and atmosphere. It affects the global temperature and climate, making the process an essential part of life. One stage of the carbon cycle is the decomposition of organic matter, in which carbon is released back into the atmosphere or soil to then be recycled for other uses.

Fungi are eukaryotic organisms that include yeast, molds, and mushrooms. They play an integral role in the carbon cycle, decomposing ground litter and woody fibers. In order to determine the efficacy of fungi in the decomposition of matter, researchers would like to better understand the relationship between different fungal species and the rates of decomposition of wood. In this project, two traits of fungi are examined: the growth rate and the moisture tolerance. Our goal is to model how different fungal species interact and decompose wood in a specific area.

We want to see how the rate of decomposition and competition between fungi is affected by different environmental changes such as moisture and seasonal fluctuations.

One fungal trait used to examine these interactions is the hyphal extension rate, which measures the growth rate of fungi. A larger hyphal extension rate for fungi corresponds to a faster growth rate. Another fungal trait we consider is the moisture tolerance of fungi, which is measured by the moisture niche width and competitive ranking. The moisture niche width is the range of moisture levels where a species is able to live and function. We define species with a greater niche width (tolerant to a wider range of moisture conditions) as generalists, and species with a narrower niche width (tolerant to a smaller range of moisture conditions) as specialists [1]. In addition to the niche width, the competitive ranking is also a factor in calculating the moisture tolerance, as it determines how a fungal species competes with other fungal species in pairwise tests and similar conditions. The moisture tolerance is calculated by taking the difference between the moisture niche width and competitive ranking, after scaling both to [0,1]. Previous research suggests that fungi with larger hyphal extension rates decompose wood faster. Additionally, fungi with slower growth rates are found to be more tolerant to environmental changes such as moisture and temperature, while fungi with faster growth rates are not as tolerant to the same changes [2].

In order to determine the decomposition of woody fibers, the loss of mass is measured as wood blocks are exposed to different fungal species. Wood has varying stages of decay, so to simplify this model we examine the middle stage of the decay cycle and assume that the results obtained are consistent for the other stages. For the model, we consider how wood decomposition is affected by an individual species' moisture tolerance and growth rate, fungal population growth in the presence of interspecies competition, and environmental moisture changes based on regional climate.

2 Principle

2.1 Niche Width & Moisture Tolerance

In evolutionary theory and ecology, it has been studied in a myriad of species that there is a set of characteristics and features that lend to the species' survival that make it particularly suited for its ecosystem or dynamic biological system. This niche dimension, defined by the ecologist G.E. Hutchinson, can be represented in n-environmental, distinct axes to characterize the population growth rate, and the fundamental niche is the portion of the niche space where the species population can persist without crashing into extinction, where

$$\mathscr{R}(\mathbf{x}) \geq 0,\ \mathbf{x} \in \mathbb{R}^n$$

where $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ denotes the vector of environmental states [3].

However, it is extremely challenging to measure and model a particular population's fitness directly across the full range of environmental conditions and its encounters. As a result, we assume that population persistence can be modeled by a small set, but ecologically relevant number of conditions, the notion of niche breadth, also known as niche width by some ecologists, allows for the quantitative measurement of a population's niche without regarding the relative frequencies of the various resources that can be accessed or limited to the organism in the environment [4]. Several assumptions about the organism's interactions with others and its environment in order to effectively model niche breadth, where if two populations have access to the same resource, then the population as a group will use such a resource in proportion to its availability [5]. Furthermore, it can be assumed that the environmental conditions are inherently a finite collection of categories, where niche breadth can be observed via measuring the distribution of organisms based on such resource states [5] [6].

To measure niche breadth, we use Hurlbert's index for distinct resource states, extending Levins' index by accounting for the unequal distribution of distinct resource states, as follows in Equation (1)

$$B = \frac{1}{\sum_{j=1}^{k} \left(\frac{p_j^2}{a_j} \right)} \tag{1}$$

where in this setting with fungi and their resource states, p_j denotes the proportions of fungal abundance observed in moisture state j, a_j denotes the proportion of the total available resource consisting of that moisture state j, and B denotes the Hurlbert's niche breadth [6] [7]. However, since B ranges from $[\frac{1}{n}, 1.0]$, we seek to normalize the niche breadth measure, we use the standardized Hurlbert's niche breadth shown below in Equation (2):

$$\widetilde{B} = \frac{B - \min(a_j)}{1 - \min(a_j)} \tag{2}$$

where $\min(a_j)$ is the smallest observed proportion of all the resources and $\widetilde{B}$ is the standardized Hurlbert's niche breadth index where $\widetilde{B} \in [0, 1]$ [8].

It can be observed that if the proportion of p_j is high in a small number of resource states a_j, this would force $\sum_{j=1}^{k} \frac{p_j^2}{a_j}$ to become relatively large. Since $\widetilde{B}$ is a strictly increasing transformation of B and B is inversely proportional to the sum, $\widetilde{B}$ would become small, approaching 0. However, suppose p_j is approximately normal across all resource states, where $p_j = \frac{1}{k}$, $\forall j$, then $\widetilde{B}$ would become become larger, approaching 1. In relation to fungi and their moisture state conditions, a larger $\widetilde{B}$ would imply that a fungal species would persist across a range of moisture conditions, making them biological generalists. In contrast, a lower $\widetilde{B}$ would suggest that the fungal species is highly adapted to only a specific, narrow subset of a_j and would not thrive in other types of moisture conditions, making them as biological specialists.

In addition to niche breadth, competitive ranking is another component in defining moisture tolerance in fungi. Competitive ranking is a measure of the ability of one species population to out-compete other species in pairwise comparisons, assuming that they were held to similar environments. These rankings can be inferred from pairwise competitive outcomes to develop multispecies hierarchies, which can be incorporated our dynamical model for fungi [9].

To simulate fungal competitive ranking, we use evolutionary game theory where the outcome for one species is dependent on what other species it interacts with. In this case, each fungal species is a "player," and each pairwise interaction between fungi reflects competitive ability. First, we make the assumption that during this competitive ranking that there is no intra-species competition and it must be one species as a collective against another during evaluation. Additionally, we assume that these contests can occur under similar conditions, and there cannot be ties between these fungal species. Let there be n fungal species in the ecosystem of interest and the payoff matrix can be written as follows below:

$$\Pi = \begin{pmatrix} \pi_{11} & \pi_{12} & \dots & \pi_{1n} \\ \pi_{21} & \pi_{22} & \dots & \pi_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \pi_{n1} & \pi_{n2} & \dots & \pi_{nn} \end{pmatrix} \tag{3}$$

where $\Pi \in \mathbb{R}^{n \times n}$ representing the payoff matrices, π_{ij} represents the payoff gained by fungus i competing against fungus j. Additionally, $\pi_{ii} = 0$ since we assumed that intraspecies competition does not occur. Thus, π_{ij} represents the probability that fungi i outcompetes fungi j, indicating larger values of π_{ij} corresponds to the larger probability of success when you have i competing against j. As a consequence, it can be deduced that $\pi_{ij} \in [0, 1]$ and $\pi_{ij} + \pi_{ji} = 1$ following the definition of probability [10].

Let X_i denote the discrete random variable where the fungi i encounters another species that is not itself is uniformly distributed,

$$X_i \sim \text{Unif}([1, n-1])$$

The probability of selection of opponent j is

$$\mathbb{P}(X_i = j) = \frac{1}{n-1}, \text{ where } j \neq i$$

Furthermore, the expectation operator $\mathbb{E}[\,\cdot\,]$ for a discrete random variable X, is defined as,

$$\mathbb{E}[X] = \sum_{i=1}^{n} x_i p_i$$

Since we are interested in the payoff of fungus i against an opponent X_j, then we can denote the parameter C_i, the competitive ranking parameter, as the discrete expectation of fungus i against a randomly selected opponent, with the Equation (4).

$$C_i := \mathbb{E}[\pi_{iX_j}] = \sum_{i=1}^{n} \pi_{ij} \mathbb{P}(X_i = j) \text{ where } \pi_{ii} = 0 \tag{4}$$

Equation (4) can be simplified as follows:

$$\Rightarrow C_i = \frac{1}{n-1} \sum_{i=1}^{n} \pi_{ij} \text{ where } \pi_{ii} = 0 \tag{5}$$

With the definition of competitive ranking and niche breadth index, we propose the formula for moisture tolerance of a fungal population i, where moisture tolerance is the ability for a fungus to survive, grow, and reproduce in a range and fluctuation of water levels. Let T_i be defined as the moisture tolerance of a fungi species i. The equation is as follows:

$$T_i = C_i - \widetilde{B}_i \tag{6}$$

where T_i reflects the difference between the competitive ranking of fungus i, C_i, and the normalized Hurlbert's niche breadth index, $\widetilde{B}$. It is important to note that since $C_i \in [0, 1]$ as well as $\widetilde{B} \in [0, 1]$, the interval of T_i is $[-1, 1]$. As a consequence, if $T_i > 0$, then the fungal species is much more adept at competing against its fungal opponents than surviving in its moisture states. If $T_i < 0$, then the fungal species is more adept at surviving in various moisture states than competing against its fungal species. Finally, if $T_i \approx 0$, then the fungal species has balanced the two.

2.2 Moisture Tolerance & Decomposition Rate

Having established the moisture tolerance index $T_i = C_i - \widetilde{B}_i$ in Equation (6), we now link this scalar characterization of each fungal species to the physical process it influences: the decomposition of woody biomass. We construct a system of ordinary differential equations (ODEs) that simultaneously tracks the total remaining woody biomass $W(t)$ and the population abundance $N_i(t)$ of each fungal species i present in the patch. Coupling these two processes captures the essential ecological feedback: fungi consume substrate to grow, and the available substrate constrains how large the fungal community can become.

Let $W(t) \geq 0$ denote the total woody biomass (in grams of dry mass) remaining in a fixed patch at time t (measured in days). Decomposition is modeled as a loss process driven by the aggregate activity of all n fungal species present:

$$\frac{dW}{dt} = -W(t) \cdot \sum_{i=1}^{n} \delta_i\big(g_i,\, T_i,\, M(t)\big) \cdot N_i(t) \tag{7}$$

where $W(t) \geq 0$ is the woody biomass (grams of dry mass) remaining at time t (days), $N_i(t)$ is the population abundance of fungal species i, $M(t) \in [0, 1]$ is the environmental moisture level at time t (its explicit formulation is discussed further below), and δ_i is the species-specific decomposition rate defined in the next section.

The negative sign in Equation (7) reflects that woody biomass is strictly lost through decomposition because it cannot spontaneously regenerate. Since $W(t) \geq 0$, $\delta_i \geq 0$, and $N_i(t) \geq 0$, the entire sum is non-negative, and the negative sign ensures $\frac{dW}{dt} \leq 0$ at all times. Furthermore, when $W(t) = 0$, the right-hand side vanishes exactly, so decomposition naturally ceases once all substrate is exhausted, and no additional boundary condition is required.

The multiplicative structure $W(t) \cdot N_i(t)$ in Equation (7) is intentional. If either the remaining substrate (wood available for decomposition) or the fungal population approaches zero, decomposition by that species effectively stops, which is ecologically consistent. When substrate is abundant, the rate of mass loss increases with the number of active decomposers present. As the substrate becomes depleted, however, decomposition slows regardless of the competitive strength of the species. This captures the ecological reality that fungal activity becomes substrate-limited during the later stages of wood decay. [2]

2.3 Species-Specific Decomposition Rate.

The term $\delta_i(g_i, T_i, M(t))$ quantifies how effectively species i decomposes biomass under current conditions. Since the problem identifies two independent trait axes, which are growth rate and moisture tolerance, we model δ_i as a product of a baseline rate and two dimensionless modifier functions, one for each trait:

$$\delta_i\big(g_i,\, T_i,\, M(t)\big) = \alpha \cdot \beta(g_i) \cdot \gamma(T_i, M(t)) \tag{8}$$

where $\alpha > 0$ is a baseline scaling constant (units: proportion mass loss per day per unit population). This parameter represents the baseline decomposition rate when fungal traits are average and environmental conditions are neutral. The multiplicative form assumes that growth rate and moisture response contribute independently to decomposition output. This assumption is consistent with the problem statement, which presents the relationships between decomposition rate and each variable separately in Figures 1 and 2 without indicating any interaction between them.

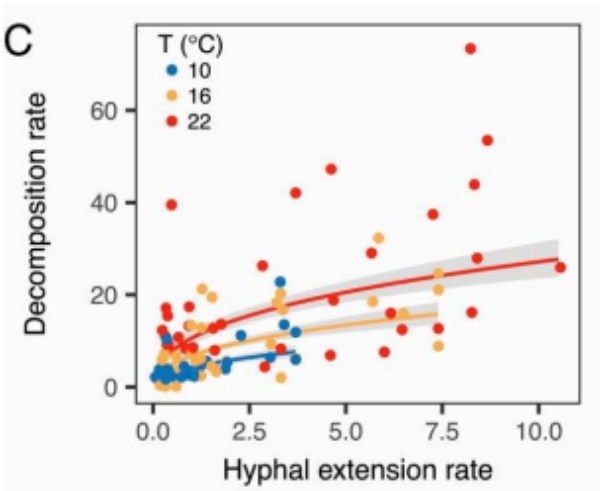

Figure 1: The relationship between the hyphal **extension rate** (mm/day) of various fungi and the resulting wood decomposition rate (% mass loss over 122 days) at various temperatures. (Figure 1C in [1]).

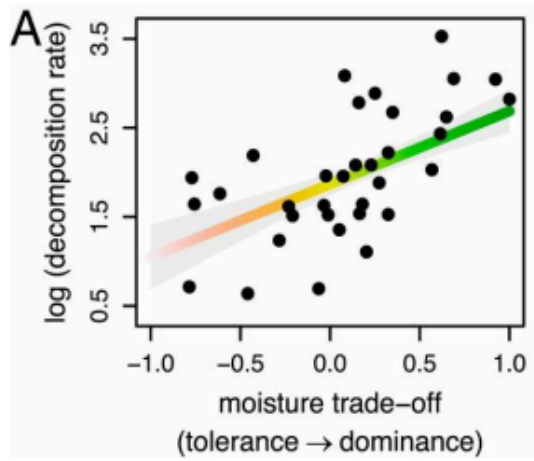

Figure 2: The relationship between the moisture tolerance (difference of each isolate's competitive ranking and their **moisture niche width**, both scaled to [0,1]) of various fungi and the resulting wood decomposition rate (% mass loss over 122 days, log transformed). (Figure 4A in [1]).

The *growth-rate modifier* $\beta(g_i)$ encodes the positive relationship between hyphal extension rate and decomposition rate shown in Fig 1. Since Fig 1 shows an approximately linear trend across temperatures, we adopt the simplest linear model:

$$\beta(g_i) = \frac{g_i}{\bar{g}} \tag{9}$$

where $\bar{g}$ is the mean hyphal extension rate across all n species in the community. Normalizing by $\bar{g}$ makes β dimensionless, with $\beta = 1$ for an average-growth species, $\beta > 1$ for fast growers, and $\beta < 1$ for slow growers. This ensures that the relative contribution of each species to decomposition scales proportionally with its growth rate, consistent with Fig 1.

The *moisture-tolerance modifier* $\gamma(T_i, M(t))$ encodes how well species i performs under the current moisture conditions. Each fungal species has an optimal moisture level at which decomposition is most efficient, with performance declining as moisture conditions move away from this optimum. We model this as a Gaussian decay:

$$\gamma\big(T_i,\, M(t)\big) = \exp\Big(-\lambda \cdot \big(M(t) - M_i^*\big)^2\Big) \tag{10}$$

where $M_i^* \in [0, 1]$ is the optimal moisture level for species i and $\lambda > 0$ controls how sharply performance falls off away from that optimum. When $M(t) = M_i^*$, we have $\gamma = 1$ and the species operates at full decomposition capacity; as $|M(t) - M_i^*|$ increases, γ decays toward 0, suppressing that species' contribution. The Gaussian form is chosen because it is symmetric around the optimum, bounded in $[0, 1]$, and consistent with the unimodal moisture-response curves commonly observed in fungal physiology [2].

The key connection to our moisture tolerance index T_i enters through M_i^*. Recall from Equation (6) that $T_i > 0$ indicates a species that is more adept at competing than at tolerating a wide range of moisture conditions, meaning that it behaves as a specialist. Such a species should perform well only near the mean environmental moisture $\bar{M}$. Conversely, $T_i < 0$ indicates a moisture generalist whose niche width is broad, meaning it can sustain decomposition across a wider range of moisture levels. We encode this by setting:

$$M_i^* = \bar{M} + \theta \cdot T_i \tag{11}$$

where $\bar{M}$ is the long-run mean moisture of the environment (parameterized per biome) and $\theta \in (0, 1)$ is a sensitivity parameter governing how strongly moisture preference shifts with T_i. It can be observed that a species with $T_i \gg 0$ will have $M_i^* > \bar{M}$, placing its optimum above the mean and making it sensitive to drier conditions. A species with $T_i \ll 0$ will have M_i^* close to or below $\bar{M}$, but the breadth of its niche means it still achieves moderate γ across many moisture states. In this way, the negative relationship between moisture tolerance and decomposition rate seen in Fig 2 emerges directly from the model structure: generalists ($T_i < 0$) sustain moderate γ across

conditions, while specialists ($T_i > 0$) achieve higher peak decomposition rates but only under narrow moisture conditions.

2.4 Fungal Population Dynamics

For the model to capture competition between species over time, the abundance $N_i(t)$ of each species must also evolve dynamically. We propose a logistic-style equation that couples population growth to both the available substrate and interspecies competition:

$$\frac{dN_i}{dt} = N_i \cdot \left[r_i \cdot \beta(g_i) \cdot \gamma(T_i, M(t)) \cdot \frac{W}{K} \; - \; \mu_i \; - \; \sum_{j \neq i} \pi_{ji} \cdot N_j \right] \tag{12}$$

where $r_i > 0$ is the intrinsic growth rate of species i, K is a carrying-capacity constant proportional to the initial biomass W_0, and $\mu_i > 0$ is the natural mortality rate. The growth term $r_i \cdot \beta(g_i) \cdot \gamma(T_i, M(t)) \cdot \frac{W}{K}$ ties population growth to the same trait modifiers used in decomposition. A species grows faster when it is well-suited to both the current moisture and has a high hyphal extension rate, but only in proportion to how much substrate remains. When $W \to 0$, growth stops for all species. The competition term $\sum_{j \neq i} \pi_{ji} N_j$ uses the payoff entries from Equations (4)–(5) directly. A species i is suppressed more when a dominant competitor j (with large π_{ji}) is abundant.

2.5 Environmental Moisture as a Forcing Function.

The environmental moisture $M(t)$ is treated as an external forcing function rather than a dynamic variable, under the assumption that moisture levels in a patch are governed by regional climate and not by fungal activity itself. To capture both the mean conditions of a biome and the temporal variability that the problem asks us to examine, we model $M(t)$ as:

$$M(t) = \bar{M} + A \sin\left(\frac{2\pi t}{P}\right) + \sigma\, \xi(t) \tag{13}$$

where $\bar{M} \in [0, 1]$ is the long-run mean moisture of the environment where higher values correspond to wetter biomes (ranging from arid to tropical rain forest), $A \geq 0$ and $P > 0$ are the amplitude and period of seasonal fluctuation, $\sigma \geq 0$ controls the severity of stochastic variability, and $\xi(t) \sim \mathcal{N}(0, 1)$ is a standard normal random variable representing purely random day-to-day fluctuations in moisture, such as unexpected rainfall or drought, that cannot be predicted by the seasonal cycle alone.

Here $\xi(t)$ is re-drawn at each time step independently, meaning the randomness has no memory of previous days. Since $\xi(t)$ has zero mean, it is equally likely to be positive or negative at any given time, so the random fluctuations cancel out on average and the long-run mean moisture remains $\bar{M}$ regardless of how large σ is. It is only through the product $\sigma\, \xi(t)$ that the randomness has any real impact on $M(t)$. When $\sigma = 0$, the random term vanishes entirely and $M(t)$ follows a purely deterministic seasonal cycle. Increasing σ introduces more severe fluctuations, allowing us to probe which species and combinations are most robust to environmental variability, directly addressing the sensitivity analysis required by the problem.

The sin term $A \sin\left(\frac{2\pi t}{P}\right)$ is chosen to model seasonal variation in moisture, as sine functions are periodic and naturally capture the recurring wet and dry cycles observed across all biomes. The period P can be set to 365 days to reflect annual seasonality, and the amplitude A controls how strongly moisture deviates from the mean $\bar{M}$ over the course of a cycle.

2.6 The Coupled System.

Together, Equations (7) and (12) form a coupled $(n+1)$-dimensional ODE system:

$$\begin{cases} \dfrac{dW}{dt} = -W \displaystyle\sum_{i=1}^{n} \alpha \cdot \beta(g_i) \cdot \gamma\big(T_i, M(t)\big) \cdot N_i \\ \dfrac{dN_i}{dt} = N_i \left[r_i \cdot \beta(g_i) \cdot \gamma\big(T_i, M(t)\big) \cdot \dfrac{W}{K} - \mu_i - \displaystyle\sum_{j \neq i} \pi_{ji} N_j \right], \quad i = 1, \ldots, n \end{cases} \tag{14}$$

with initial conditions $W(0) = W_0$ and $N_i(0) = N_{i,0}$. This system is the central object of our analysis in Section 3.

3 Results

3.1 Single Fungal Species Decomposition Trajectory

We first consider the simplest version of the model, in which only one fungal species is present. In this case, the interspecific competition term is removed, so the system reduces to

$$\frac{dW}{dt} = -W(t)\,\delta(g, T, M(t))\,N(t),$$

$$\frac{dN}{dt} = N(t)\left[r\,\beta(g)\,\gamma(T, M(t))\frac{W}{K} - \mu\right].$$

This baseline case isolates the interaction between fungal growth and substrate depletion before introducing additional ecological complexity such as pairwise competition or diversity effects.

Figure 1 shows the time evolution of wood biomass $W(t)$. The trajectory decreases monotonically over time, which is consistent with the structure of the model since fungal activity can only reduce the amount of wood remaining. The decline is steeper during the earlier part of the simulation, when both the fungal population and the available substrate are sufficiently large. As the wood biomass becomes depleted, the rate of decomposition gradually slows.

Figure 2 shows the corresponding fungal abundance trajectory $N(t)$. The fungal population initially increases because substrate is abundant and environmental moisture remains close enough to the species' preferred range to support growth. However, this increase does not continue indefinitely. As decomposition proceeds, the wood biomass decreases, which weakens the substrate-dependent growth term W/K. As a result, the fungal abundance eventually levels off and begins to decline.

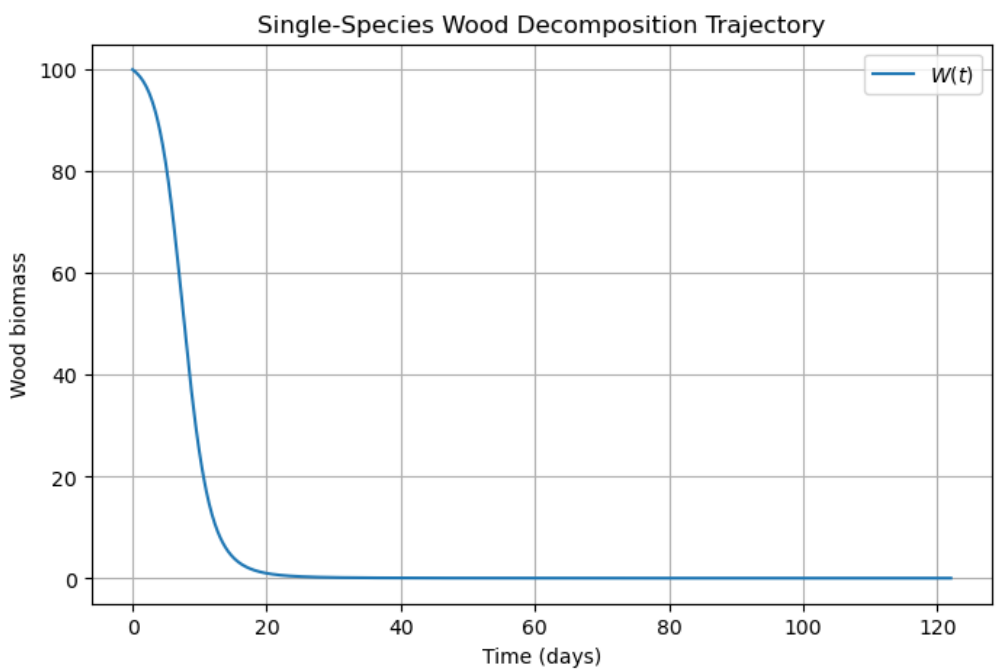

Fig 1. Trajectory of wood biomass $W(t)$ in the baseline single-species simulation. The wood mass decreases monotonically over time, with faster decomposition occurring earlier when fungal activity and substrate availability are both relatively high.

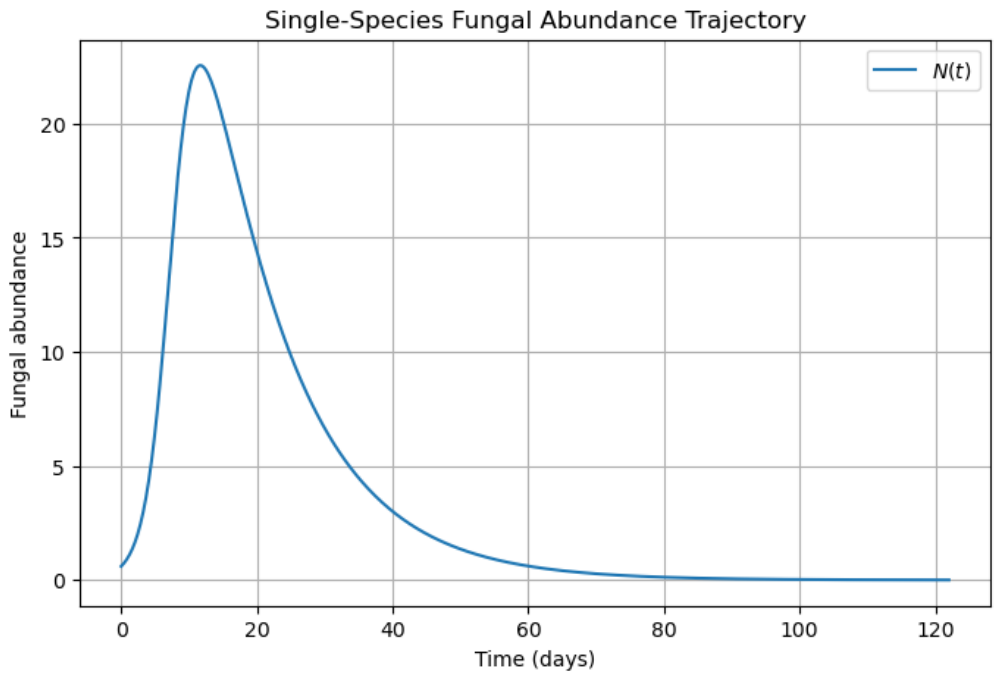

Fig 2. Trajectory of fungal abundance $N(t)$ in the baseline single-species simulation. The fungal population initially grows under favorable substrate conditions, then slows and declines as the wood resource becomes depleted.

3.2 Parameter Sweep for Moisture Sensitivity Across Biomes

To study how environmental moisture affects decomposition, we next compare the single-species model across several biome conditions. In this section, the fungal trait parameters are held fixed, while the long-run mean environmental moisture $\bar{M}$ is varied to represent different biome types. We consider three cases: a dry biome, a moderate biome, and a wet biome.

The purpose of this parameter sweep is to isolate the effect of environmental moisture on decomposition. Since the fungal species is kept fixed, any differences in the resulting trajectories are caused by changes in the moisture forcing rather than changes in fungal traits. This allows us to evaluate how sensitive the model is to biome-level moisture conditions.

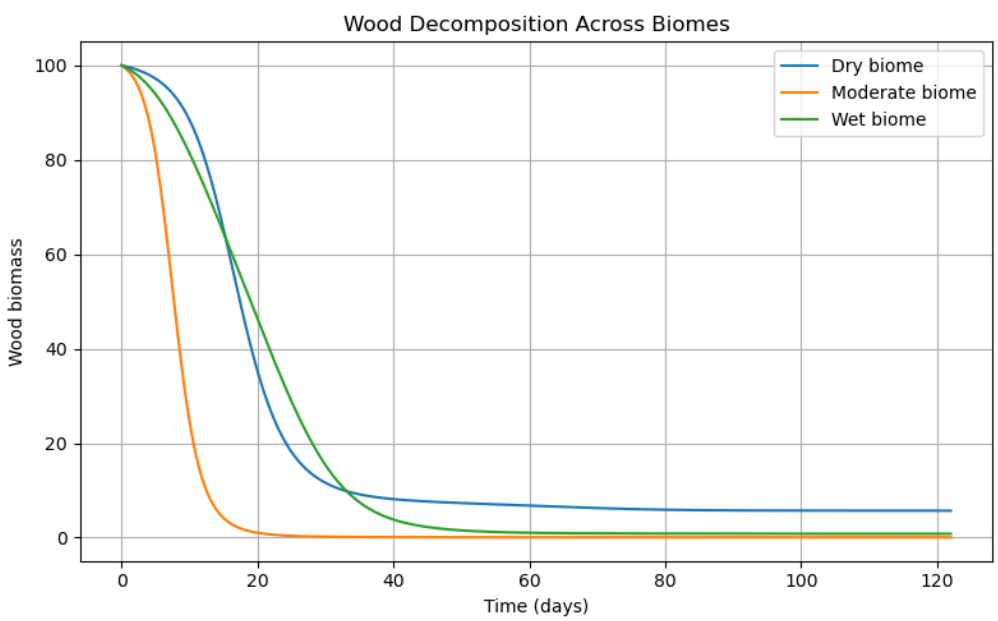

Fig 3. Wood biomass trajectories $W(t)$ under dry, moderate, and wet biome moisture conditions. Decomposition proceeds fastest when the environmental moisture is closest to the fungal species' preferred moisture level.

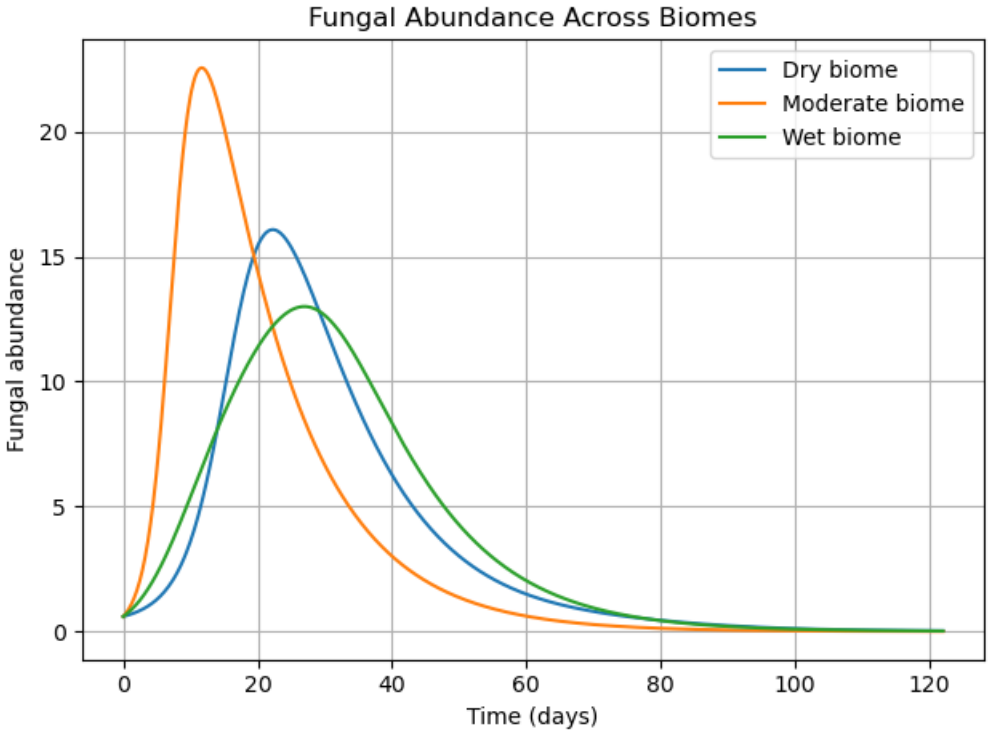

Fig 4. Fungal abundance trajectories $N(t)$ under dry, moderate, and wet biome moisture conditions. Fungal growth is strongest when the environmental moisture is closest to the species' preferred moisture level, leading to a higher and earlier population peak in the moderate biome case.

Overall, these results show that the model is highly sensitive to mean moisture conditions. Even with the same fungal species and the same initial wood biomass, changing the biome moisture level can substantially alter both fungal abundance and the decomposition trajectory. This supports the ecological interpretation that fungal performance depends not only on intrinsic growth rate, but also on how well the environment matches each species' moisture preference.

3.3 Seasonal Forcing Analysis and Stochastic Robustness

We next examine how temporal variability in environmental moisture affects decomposition in the single-species model. In this section, the fungal traits are held fixed while the seasonal forcing amplitude A and the stochastic noise level σ are varied. This allows us to study the sensitivity of the decomposition trajectory to both periodic and random environmental fluctuations.

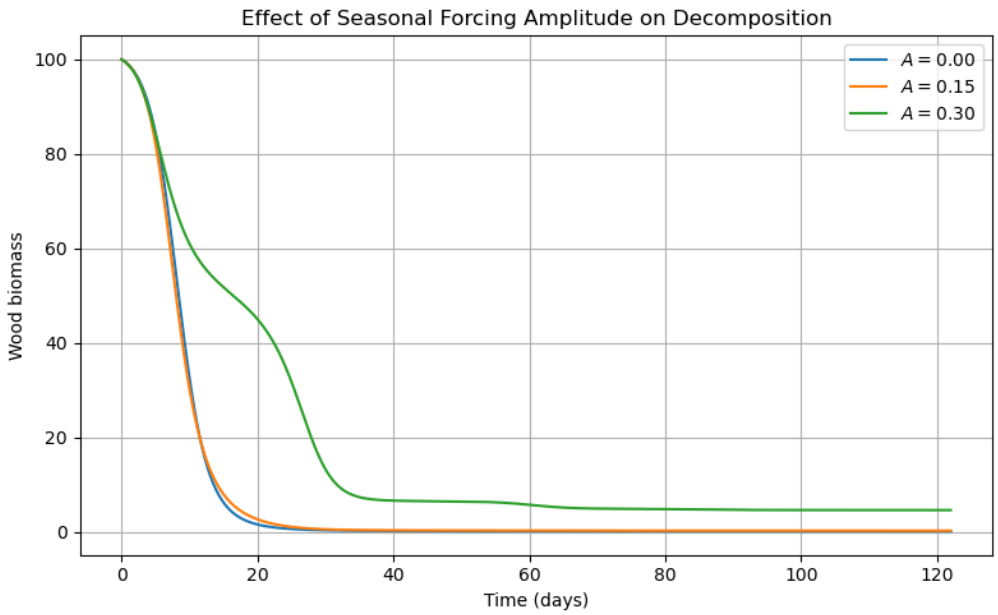

Fig 5. Wood biomass trajectories $W(t)$ under different seasonal forcing amplitudes A. Larger periodic moisture oscillations push the environment farther from the fungal species' preferred moisture level and slow decomposition.

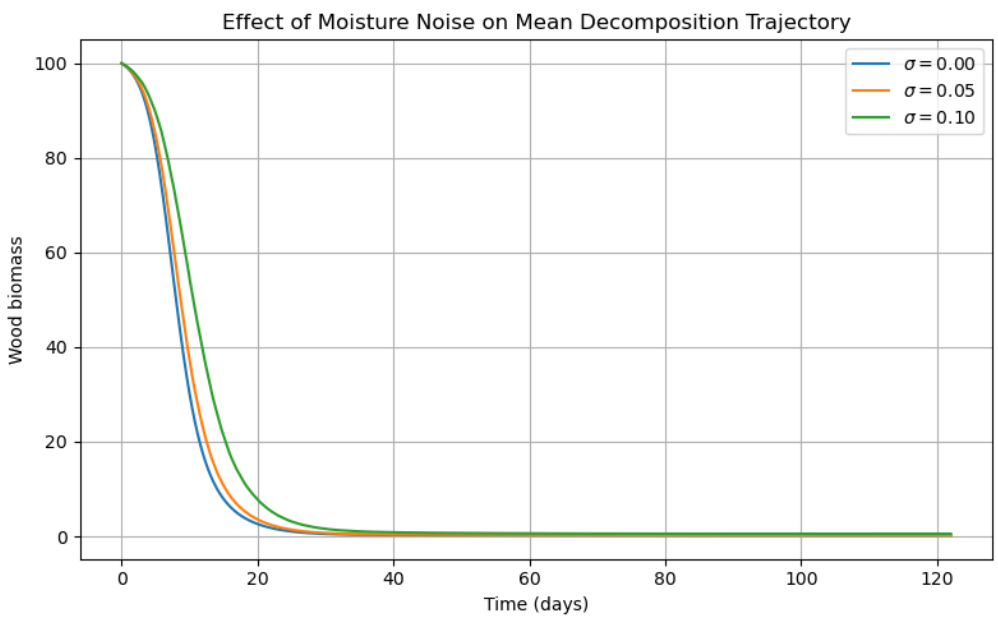

Fig 6. Mean wood biomass trajectories $W(t)$ under different stochastic noise levels σ, averaged over multiple simulation runs. Stronger random moisture fluctuations reduce decomposition efficiency on average, while preserving the overall monotone decay pattern.

3.4 Pairwise Competition of Fungal Species

In this section, we seek to simulate the fully-coupled dynamic ordinary differential equation system (14) for 2 species, where,

$$
\begin{cases}
\dfrac{dW}{dt} = -W \displaystyle\sum_{i=1}^{2} \alpha\, \beta(g_i)\, \gamma(T_i, M(t))\, N_i \\[2ex]
\dfrac{dN_i}{dt} = N_i \left[r_i\, \beta(g_i)\, \gamma(T_i, M(t))\, \dfrac{W}{K} - \mu_i - \displaystyle\sum_{j \neq i} \pi_{ji} N_j \right], \; i = 1, 2
\end{cases}
\tag{15}
$$

We have reduced this model to a two-species case, sharing a single wood substrate to understand and observe how the coupled system (14) resolves and implements interspecies competition as highlighted in previous literature. Hence, this $n = 2$ species isolates the interaction between the growth-rate modifier $\beta(g_i)$ (9) (12) and the pairwise competition term $\sum_{j \neq i} \pi_{ji} N_j$ (12). Although this assumes that there are no higher-order and more complex community dynamics between multiple fungal species, we opt to trace the relationship between trait differences and competitive outcomes instead of these global interaction systems.

We construct a species pair that represents the growth-competition tradeoff as demonstrated in previous literature. Species 1 is a fast-growing specialist with $g_1 = 7.0$mm/day and $\widetilde{B}_{e,1} = 0.3$ with a weaker competitive capacity $\pi_{1,2} = 0.40$. In contrast, species 2 is a slow-growing generalist $g_2 = 3.0$mm/day and $\widetilde{B}_{e,2} = 0.7$ with a stronger competitive capacity $\pi_{2,1} = 0.60$. Computing the moisture tolerance equation (6) yields $T_1 > 0$ and $T_2 < 0$ for species 1 and species 2, respectively, emblematic of the opposite ends of the specialist-generalist paradigm. Under constant moderate moisture with a deterministic seasonal pattern ($\bar{M} = 0.5,\ \sigma = 0$), species 1 will benefit from substrate-driven growth while species 2 is favored by a higher-payoff symmetry. For this section, we assume that the payoff matrices are fixed. We will explore a dynamically changing payoff system in the next section. We first simulate baseline dynamics from symmetric initial conditions over 300 days and then vary the initial conditions afterwards.

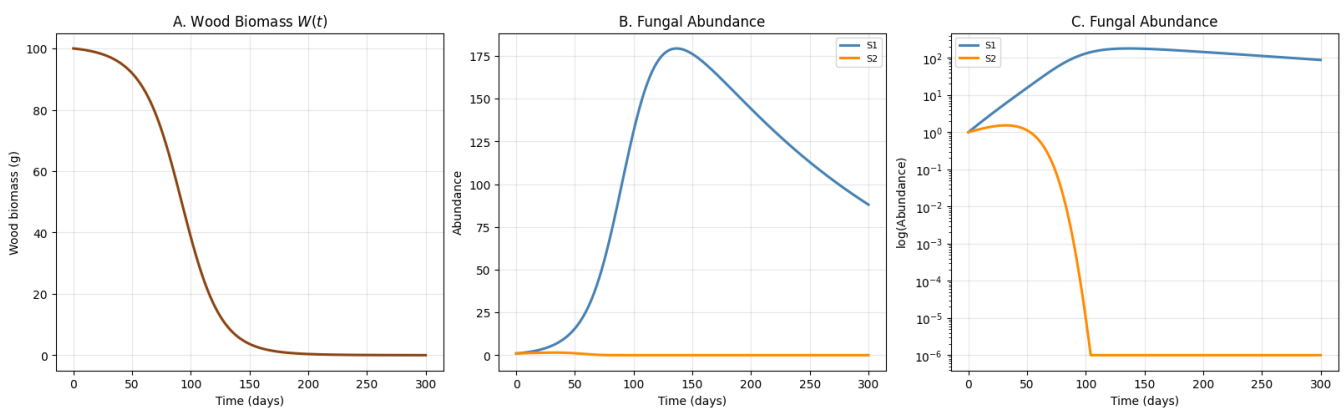

Fig 7.
Figure 7.A. Wood biomass $W(t)$ over 300 days during the interaction between two species 1 and 2. **Figure 7.B.** A linear scale representation of the fungal abundance between species 1 and 2. **Figure 7.C.** For a clearer view, a semi-log plot of fungal abundance between 1 and 2.

In 7, we observe that the fungal abundance of Species 1 persists over 300 days while there is total collapse of Species 2. Although there is an initial incline of Species 2 in terms of its abundance, the 7 C highlights the quick and sharp decline of species 2's collapse. This simulation of previous literature highlights the mismatch between the growth and competition, where the competition term is minuscule compared to the growth term $r_i\, \beta(g_i)\, \gamma(T_i, M(T_i)) W/K$. Thus, even though species 1 is a weaker competitor compared to species 2, its growth capability surpasses that of species 2's growth rate and competition term. The payoff parameter π_{12} sweep conducted in 9 validates the work here. While there are variations in both species 1 and species 2 abundance, where species 2 has a wider variance of fungal abundance than species 1, the final trajectories are the same. It is interesting to note, however, that the stability of species 1 and species 2 is dependent on their initial condition of abundance. Although the three separate initial value conditions of $N_1(0)$ and $N_2(0)$ led to the collapse of the species 2 fungal population, species 1's collapse resulted when its initial condition was relatively too low compared to the species 2 fungal population where $N_1(0) = 1,\ N_2(0) = 20$, causing species 2 to be re-established. This implies that there exists some bistability, due to initial conditions and parameters, that switches species 1 to collapse while species 2 persists and vice versa.

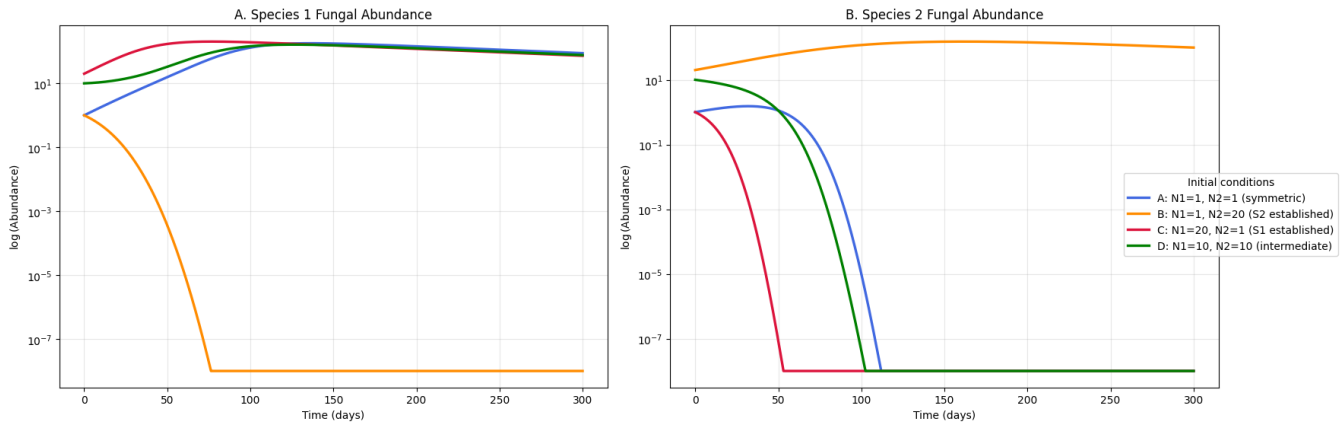

Fig 8.
Figure 8.A, 8.B Simulated trajectories of fungal abudance between species 1 **8.A.** and species 2 **8.B.**, respectively shown in a semi-log scale. π_{ji} is fixed as highlighted above. Differences in initial conditions demonstrate significant divergence in long-term stability, influencing which species persists or collapses.

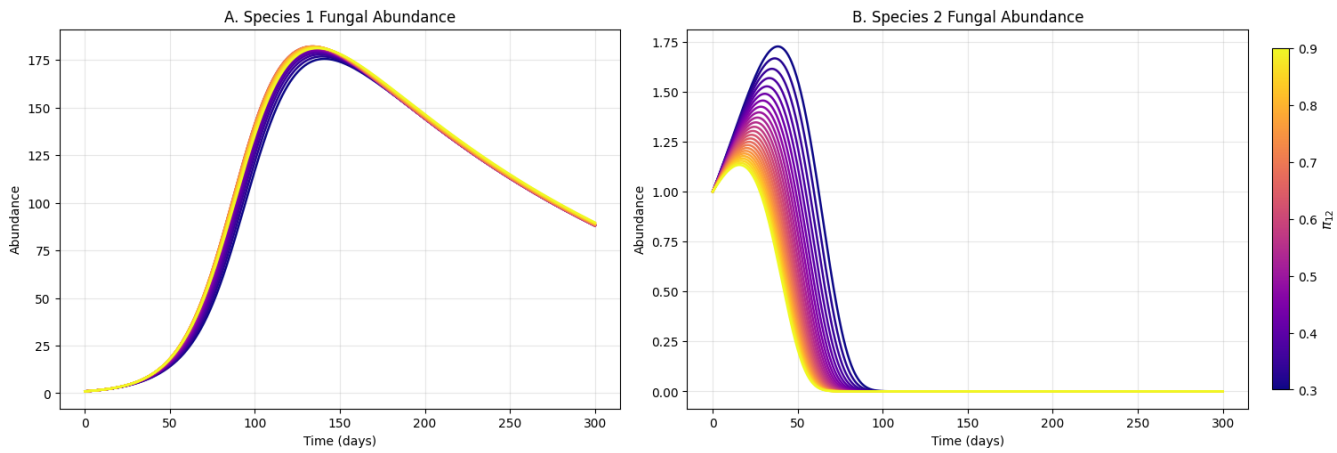

Fig 9.
Figure 9.A., 9.B. Simulated trajectories of π_{12} parameter sweep between Species 1 **9.A.** and **9.B.** from values between 0.3 to 0.9. Differences in π_{12} show varied shifts in species shifts over time, especially creating higher variance in species 2 than species 1.

3.5 Elo-Based Systems for Dynamically Changing Competitive Rankings

In Section 3.4, we examined a coupled ordinary differential equation for a fixed π_{ij} which resolves the competitive outcome underneath fixed environmental conditions, allowing for direct observation of species collapse versus persistence based on a single wood substrate. However, with n species and being $\binom{n}{2}$ possible unique interspecies interactions, section 3.4 does not account for a dynamic hierarchy between various species. For example, let species i dominate species j. Then, species j dominates species k. However, this does not necessarily imply that species i dominates species k if species k possesses a traits that could overwhelm or be more efficient than that of species i. In order to address this intransitive pairwise competition tournament system, we utilize the Elo rating system, originally developed for chess by Hungarian-American chessmaster and physicist Arpad Elo.

For each pairwise matchup, between i and j, the observed outcome is denoted as S_{ij}. There are several caveats to S_{ij} as a discrete random variable, however. Assuming that the moisture forcing term is deterministic where $\sigma = 0$, then for every fixed initial condition and parameter, the ODE produces the same trajectory for every trial, meaning that S_{ij} is a Bernoulli random variable with $p \in \{0, 1\}$ with complete certainty. This discrete random variable, becomes meaningful if the stochastic moisture forcing term is $\sigma > 0$, since each trial of the coupled ODE results in a different moisture path, indicating that each competitive outcome could be determined between trials. Therefore, over many IID trials, species i will win some fraction p_{ij} of the time, so the individual outcome will be,

$$S_{ij} \sim \text{Bernoulli}(p_{ij})$$

where, $p_{ij} \in [0,1]$ is the probability that species i overcomes species j's abundance under the environmental conditions. Again, to satisfy the definition of a Bernoulli random variable, $p_{ij} + p_{ji} = 1$, but $p_{ij} \neq \pi_{ij}$ since π_{ij} only reflects the fungal species' intrinsic competitive capabilities.

After each matchup and outcome, the matches are updated as follows:

$$E_i' = E_i + K(S_{ij} - \widetilde{S_{ij}}), \qquad \widetilde{S_{ij}} = \frac{1}{1 + 10^{\frac{(E_j - E_i)}{400}}} \tag{16}$$

where E_i and E_i' denotes the rating of the species i and its updated rating, respectively, $\widetilde{S_{ij}}$ represents the logistic Elo performance rating based on current rating difference and K is the attenuation factor that determines the weight that should be given to the species' performance relative to its pre-competition matchup rating [11]. The algorithm that utilizes an Elo-driven approach is provided as below:

Algorithm 1 Elo-Based Dynamic Ranking for Fungal Abundance

Require: Species pool $\{1, 2, \ldots, n\}$ with traits (g_i, T_i) for each species i
Require: Environmental parameters $(\bar{M}, A, P, \sigma)$
Require: Initial biomass W_0, initial abundances $N_i(0)\ \forall\, i$
Require: Update attenuator constant K
Require: Number of permutations R

```
for Each permutation r = 1, 2, ..., R do
    (1) Initialize E_i^(r) ← 1500 ∀i
    (2) Generate a random permutation of all unique pairs of (i, j)
    for Each pair (i, j) sequence permutation do
        (3) Solve the ODE system (14) with n = 2 species i, j
        under stochastic moisture forcing (13) over [0, t_f]
        (4) Determine the outcome S_ij
        (5) Compute expected outcomes from current ratings S̃_ij
        (6) Update ratings E_i^(r) and E_j^(r)
    end for
    (7) E_i ← E_i + E_i^(r) ∀i
end for
```

The simulation initializes five fungal species with hyphal extension rates, ranging from $g_1 = 5.5$mm/day to $g_5 = 4.5$mm/day. This narrow range ensures that no single species can dominate through growth alone, which is an intentional departure from the literature on hyphal extension rates and competition. This range also produces a narrow range of growth-rate modifiers $\beta(g_i)$ which is designed to isolate the role of competitive dynamics when intrinsic advantages are marginal. The niche breadth width $\widetilde{B}_i$ are also ranged in an interval between $[0.20, 0.80]$ which spans the specialist-generalist axis, starting with species 1 as a specialist to species 5 as a generalist. Most importantly, the payoff matrix is initialized with intransitivity where species 1 beats species 3, species 3 beats species 5, and species 5 beats species 1. The Elo ratings are initialized as $E_i = 1500 + 400 \log_{10}(C_i/(1 - C_i))$, so the initial competitive hierarchy is preserved instead of assuming that all the species are inherently uniform from start.

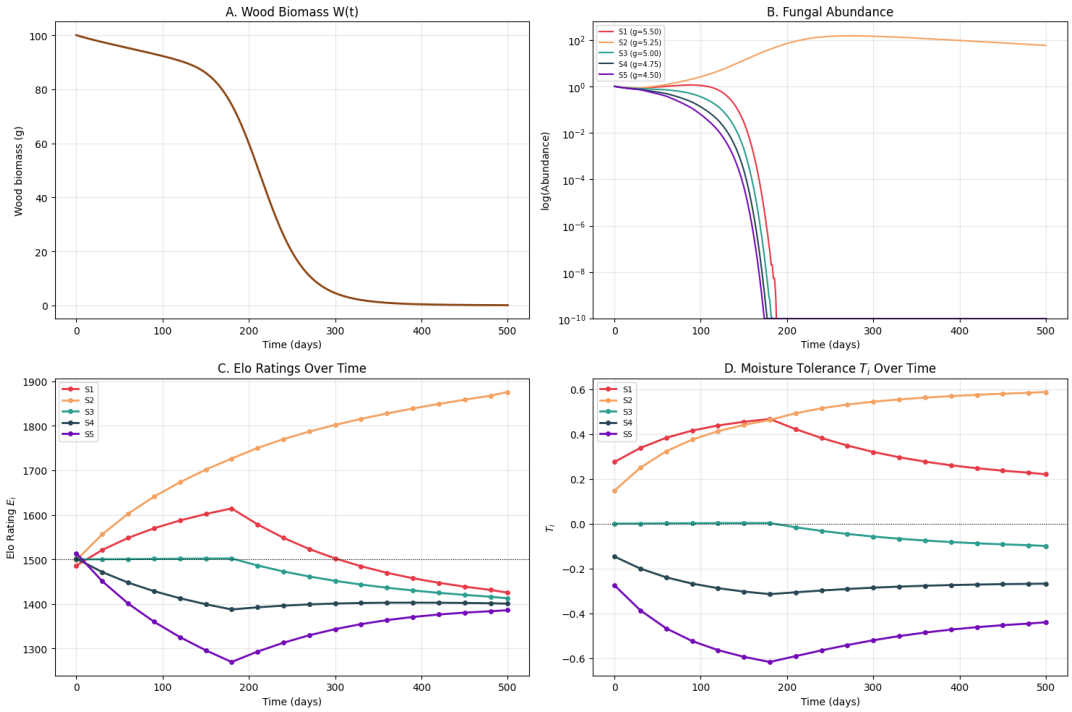

Fig 10.
Figure 10.A. Wood biomass $W(t)$ overtime affected by the interaction of 5 species.**Figure 10.B.** Fungal abundance of 5 species with different traits, calibrated to start at 1500 Elo rating and $\sigma = 0$. **Figure 10.C.** Elo ratings of each 5 species over time. **Figure 10.D.** Moisture tolerance of 5 species over time.

Under constant moisture $\bar{M} = 0.5$ with no stochastic forcing $\sigma = 0$, the deterministic simulation produces one, reproducible outcome where species 2 is the dominant competitor and all other species collapse before 250 days 10. Observing species 2, we see that it benefits from a combination of a moderate $\beta(g_2) = 1.05$ and a pairwise advantage over species 1. Although species 1 does rise due to a higher growth rate, species 1 cannot grow fast enough to exert advantage over species 2 while species 2 climbs in Elo rating.

As stated above, the deterministic outcome for a dynamically changing competition system is limiting for our analysis. Therefore, we use a Monte Carlo simulation to extend this analysis where we set stochastic moisture to $\sigma = 0.8$ and seasonal amplitude to $A = 10$ over a period of $P = 365$ days. For each of the 50 IID realizations, a complete moisture trajectory along with its Elo rating projections are generated by sampling $\xi(t) \sim \mathcal{N}(0, 1)$ independently at each time step, along with the seasonal component addressed in 13. In 11 and 12, 50 realizations shows that species 1 wins 30% of the time while species 2 wins 70% of the time, yielding a pairwise probability of $p_{21} = 0.70$ and $p_{12} = 0.30$. This is consistent with the relationship that we have established with environmental moisture and the specialist-generalist paradigm. Since species 1 is a pre-determined specialist whose optimal temperature $M_1^* = \bar{M} + \theta T_1$ is above the mean, the Gaussian decay modifier $\gamma(T_i, M(t))$ can favor species 1 over species 2 temporarily. However, if this persists long enough, at around the 100-day mark, this is long enough to completely redirect the trajectory for fungal abundance. Species 3 through 5 can never win under these parameters and/or timescale despite intransitivity, which highlights that this fungal species relationship, along with an Elo system, amplifies competitive exclusion, causing their inevitable collapse.

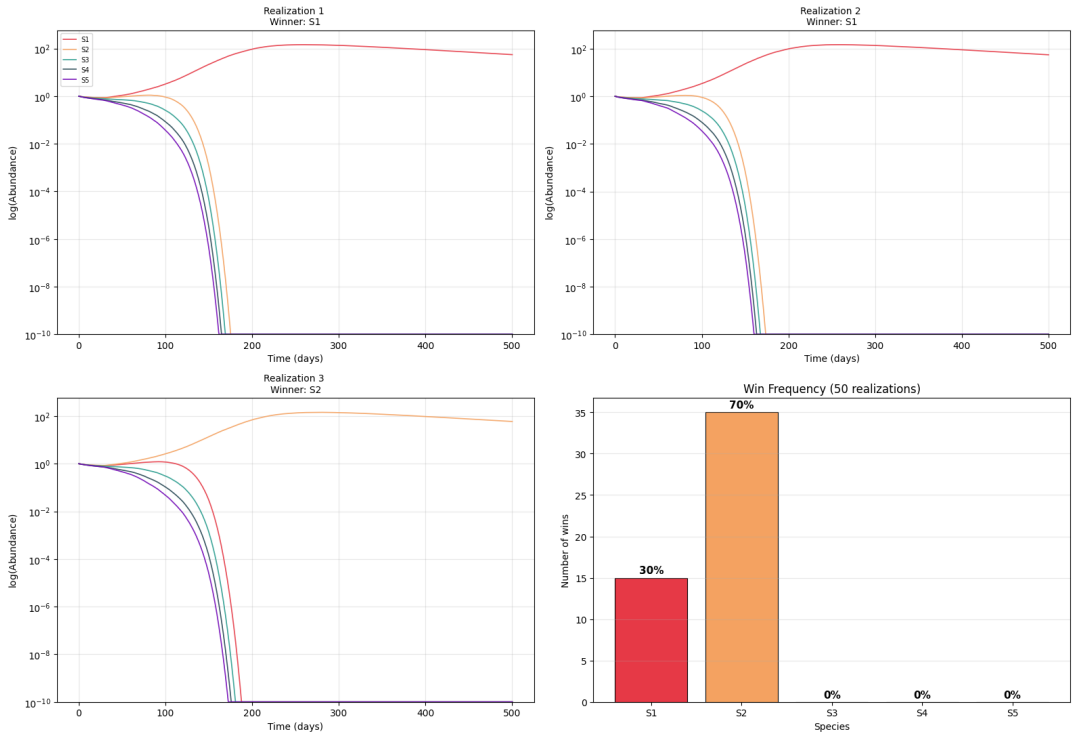

Fig 11. Figure 11.A.-C. Fungal abundance between 1 to 3 realizations after 500 days for competition between 5 species. **Figure 11.D.** Win frequency between 5 species as a bar plot after 50 realizations.

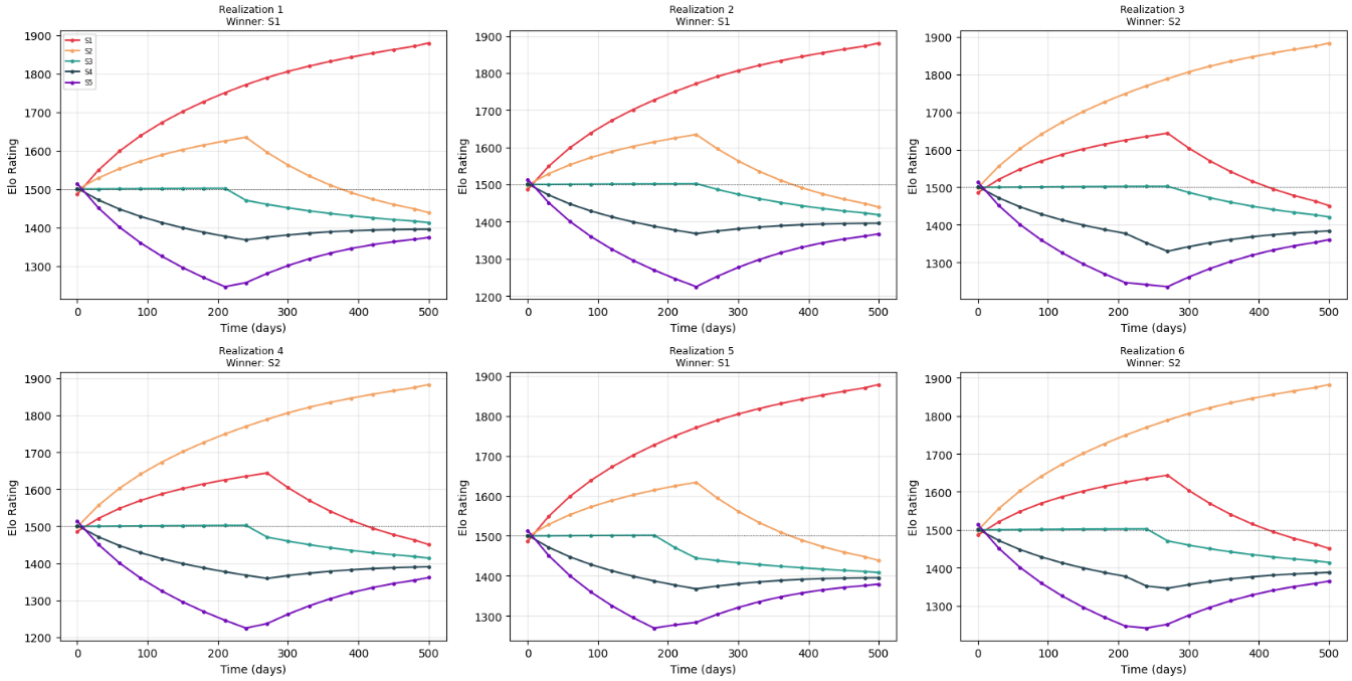

Fig 12. Elo Ratings for 6 realizations.

3.6 Relationship Between Diversity & Decomposition

To examine how the diversity of fungal communities affects the overall efficiency of wood biomass decomposition, we simulate the coupled system (14) for communities of increasing species richness: $n = 1, 2, 4$, and 8 species (geometric sequence). The single species baseline uses *Porodisculus pendulus* ($g = 2.90$ mm day^{-1}), a mid-range isolate from Table S4 of [2]. The two species community adds the fastest growing isolate *Merulius tremellosus* C3E ($g = 8.67$ mm

day^{-1}, specialist, $T_i > 0$) and the slowest growing isolate *Armillaria gallica* A4A ($g = 0.66$ mm day^{-1}, generalist, $T_i < 0$), representing the two extremes of the specialist generalist axis. The four and eight species communities add intermediate isolates from Table S4, spanning the full range of observed hyphal extension rates.

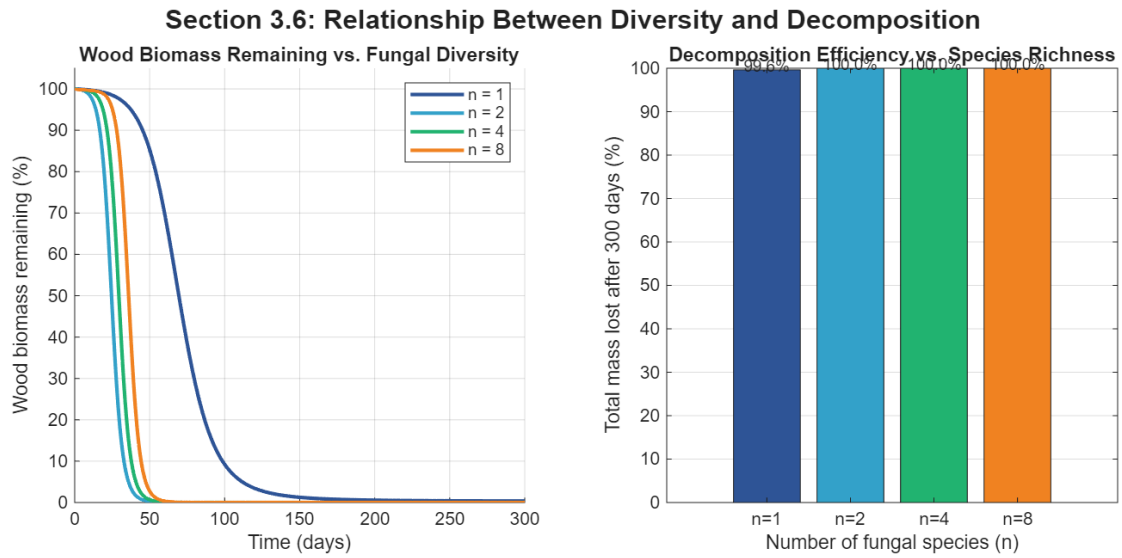

Fig 13. Section 3.6: Left: Trajectories of wood biomass remaining (%) over 300 days for communities of $n = 1, 2, 4, 8$ fungal species. Right: Total mass lost after 300 days as a function of species richness. Species parameters taken from Tables S3 and S4 of [2].

The left figure shows that increasing species richness accelerates the rate of decomposition. The single-species community ($n = 1$) decomposes the substrate the slowest, with the sigmoidal decline in biomass occurring over a longer period. As additional species are introduced, the onset of rapid decomposition shifts earlier in time, and the community collectively depletes the substrate faster. This result is consistent with the complementarity hypothesis in biodiversity-ecosystem function research. Communities containing both fast-growing specialists and moisture-tolerant generalists are able to maintain high decomposition activity across a wider range of conditions than any single species alone [2].

However, the right panel reveals a limitation of the current model. All communities ultimately reach near-complete substrate depletion ($\approx 100\%$ mass loss) by day 300, leaving little quantitative difference in total mass lost across diversity levels. This is a consequence of the long simulation window relative to the decomposition timescale. The model more meaningfully distinguishes communities by the *speed* of decomposition rather than its final extent under these parameter settings.

3.7 Perturbation Analysis and Recovery

To assess the robustness of the fungal community to sudden environmental shocks, we subject the two species system (*Merulius tremellosus* C3E and *Armillaria gallica* A4A) to abrupt changes in environmental moisture during the interval $t \in [100, 160]$ days. Three scenarios are compared: a baseline with constant $\bar{M} = 0.50$, a drought scenario in which moisture drops to $M = 0.10$, and a flood scenario in which moisture spikes to $M = 0.90$. Before and after the shock window, moisture returns to $\bar{M} = 0.50$.

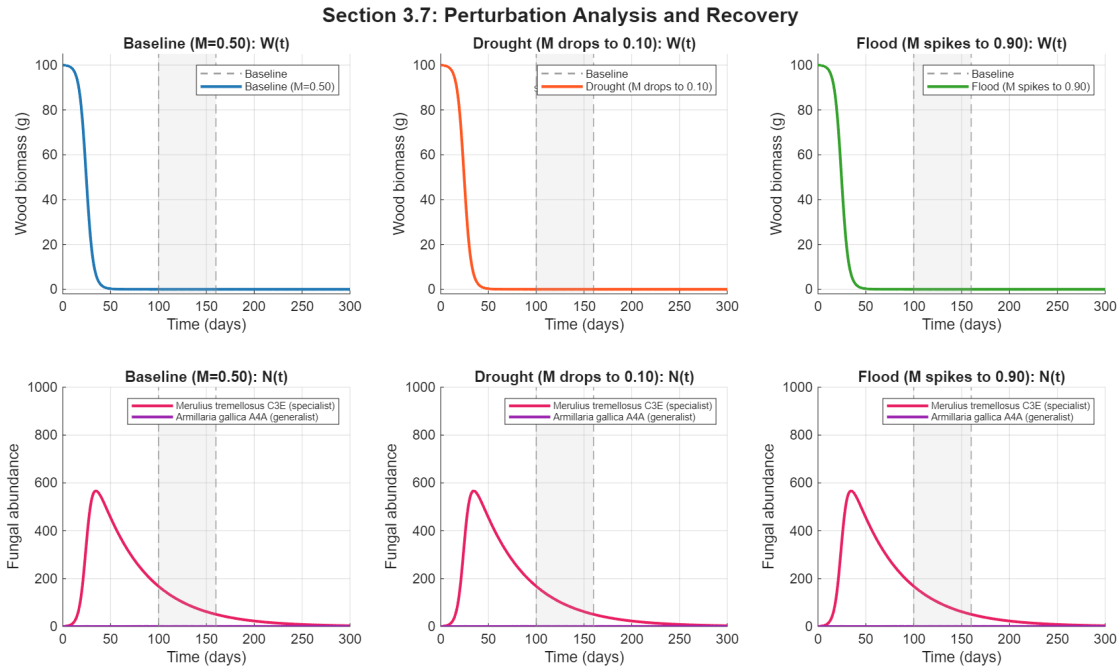

Fig 14. Section 3.7: Wood biomass $W(t)$ (top row) and fungal abundance $N(t)$ (bottom row) under baseline, drought, and flood conditions. The gray shaded band indicates the shock period ($t = 100$ to $t = 160$ days). Species parameters taken from Tables S3 and S4 of [2].

The figure shows that by $t = 100$ days, the substrate has already been substantially depleted under all three scenarios, so the shock window has limited impact on the remaining biomass trajectory. The top row shows that $W(t)$ is nearly identical across all three conditions, indicating that the perturbation arrives too late in the decomposition cycle to meaningfully alter the outcome. It suggests that once a fast growing specialist such as *Merulius tremellosus* has established and consumed the majority of the substrate, the system becomes robust to environmental shocks not because of community resilience, but because there is little remaining substrate for the shock to affect.

The bottom row reveals a second notable result: *Armillaria gallica* A4A (generalist, $T_i < 0$) remains effectively absent throughout all three scenarios. Despite its moisture tolerance advantage, the generalist is competitively excluded by the faster growing specialist under moderate baseline conditions ($\bar{M} = 0.50$), which closely matches the specialist's optimal moisture level $M_i^* = \bar{M} + \theta T_i > \bar{M}$. Neither the drought nor the flood shock is sufficient to shift conditions far enough from the specialist's optimum to allow the generalist to recover. In stable or moderately variable environments, fast growing specialists are expected to dominate and generalists are suppressed, regardless of short term moisture shocks.

4 Discussion

In the simple single-species model, we observe that wood biomass quickly decreases when the fungus is first introduced, while fungal abundance initially increases as the fungus consumes the available substrate. As the wood becomes depleted, fungal abundance also begins to decrease because of the reduced availability of substrate, showing the dependence of fungi on wood for its survival. Environmental moisture plays an important role in determining the rate of wood decomposition. Decomposition occurs most rapidly and fungal abundance is highest when environmental moisture is closest to the fungal species' preferred moisture level. As a result, the most efficient wood decomposition occurs in moderate biomes such as temperate environments. Increasing the seasonal forcing amplitude or the stochastic noise level decreases the decomposition rate because these changes create a wider range of moisture conditions. Under these circumstances, fungi spend less time in its preferred moisture conditions, which reduces its overall growth and decomposition activity.

In the two-species model, competition between species adds some complexity to the model. A fast-growing specialist with weaker competitive capacity increases in abundance, while a slow-growing generalist with stronger competitive capacity declines to extinction with most initial conditions. In this case, a higher growth rate is better for a species than a stronger competitive ability. However, when the initial fungal abundances strongly favor the

slow-growing generalist, that species can dominate the fast-growing specialist. This demonstrates that initial population conditions greatly influence the outcome of species interactions. The multi-species model using the Elo-based system also shows how interactions change over time. In the short term, specialists increase quickly because their high growth rates allow them to take advantage of favorable environmental conditions. Over longer time periods, however, species with moderate growth rates and pairwise competitive advantages may become dominant. These results show a trade-off between rapid growth under ideal environmental conditions and competitive strength under more variable conditions.

The results also suggest that biodiversity can influence decomposition efficiency. Fungal communities that contain both fast-growing specialists and moisture-tolerant generalists are able to maintain higher decomposition rates across a wider range of environmental conditions than any single species itself. Specialists increase in abundance when moisture levels are close to their optimal growth range, while generalists are able to tolerate fluctuations in moisture levels better. As a result, fungal communities containing both types of species can maintain decomposition activity even as environmental conditions change. This suggests that greater biodiversity may help stabilize wood decomposition in environments with variable moisture conditions.

One strength of the model is that it accounts for several important factors that influence wood decomposition. In particular, it accounts for resource limitation, moisture variability, and competition among fungal species. The model also explores how seasonal forcing and stochastic environmental fluctuations influence fungal growth and decomposition of wood. It demonstrates how environmental variability and species interactions affect the efficiency of wood decomposition.

However, the model also has several limitations. The model mainly differentiates fungal communities by the rate of wood decomposition rather than the total amount of wood biomass lost. In many simulations, wood biomass approaches nearly complete mass loss after a certain amount of time, making long-term differences between communities difficult to determine. As a result, the model is most useful for comparing short-term decomposition rates. Additionally, the model centers around fungal growth rate and moisture tolerance when determining decomposition rates, while other environmental factors such as temperature variation, wood type, and nutrient availability are not included. These factors can significantly influence fungal growth and decomposition processes in natural ecosystems.

Many assumptions made also influence the model's outcomes. The model assumes simplified competitive interactions among fungal species, which may not fully capture the complexity of real fungal communities. It also assumes that environmental conditions remain spatially uniform throughout the area, while real environments often have varying levels of moisture and other environmental factors within a given area.

One refinement for our model would be to apply the model to multiple specific fungal species with species-specific growth parameters and environmental preferences. This would allow us to better analyze the wood decomposition of individual species as well as multi-species communities. Another possible refinement would be to consider how other environmental variables such as temperature variations affect wood decomposition. Including these additional factors would provide a more realistic representation of environmental conditions. Finally, calibrating the model using empirical data on fungal growth rates and wood decomposition measurements could improve the model's accuracy and allow it to be applied to more specific ecosystems.

5 Conclusion

To better understand the role fungi play in decomposition, we created a model to analyze the decomposition of woody fibers within a piece of land. The model supports the idea that fungal activity and environmental conditions strongly influence wood decomposition. Moisture fluctuations decrease decomposition rates, while fungal competition can increase decomposition efficiency. Greater fungal biodiversity also improves decomposition rates, as different species have different strengths and can continue decomposing wood even under varying environmental conditions.

With these findings, we can further explore how different fungal traits contribute to or hinder wood decomposition, as well as how additional environmental factors, such as temperature, affect decomposition and interspecies competition.

The decomposition of woody fibers also has broader impacts because it releases carbon back into the soil and atmosphere, contributing to the carbon cycle which in turn influences global temperature and climate. Although this model is limited by simplifying assumptions and a restricted number of factors, future work could expand the model to incorporate additional fungal traits that increase biodiversity, as well as more environmental parameters. This would allow us to better understand how changes in decomposition at a small scale may influence larger-scale processes such as global climate change. Understanding the relationship between fungal diversity and environmental variation highlights the important role fungi play in wood decomposition and carbon release.

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